

Addis Ababa Science and Technology University

College of Social and Natural Science

Logic and Critical Thinking

Mid-Exam

Total Points: 25% Time Allowed: 60' Date: August 26, 2021

Name _____ ID.NO _____ Section _____

Part I: Write "True" if the statement is correct and write "False" if it is wrong (5pts.)

1. In all cases, strength of and inductive argument is independent of the truth values of the premise and conclusion statements of an argument.
2. A lexical definition should be narrow, instead of being broad.
3. Expository passage is one form of logically acceptable deductive arguments.
4. The purpose of a precisising definition is to reduce vagueness of the term.
5. In argumentative passage, inferential claim refers to the rational or logical relations between a premise and conclusion statements.

Part II: Choose the best answer and write your choice on the space provided (10pts.)

1. *'For the past twenty five years Ethiopian athletes have been dominant in the world of athletics. Thus, Ethiopian athletes will remain dominant in the coming couple of years.'* This argument is _____.
A. Valid deductive argument
B. Weak inductive argument
C. strong inductive argument
D. invalid inductive argument
2. Which one of the following series of terms are listed in the order of decreasing intension?
A. Ethiopia, Africa, Adama, World
B. Africa, World, Adama, Ethiopia
C. Adama, Ethiopia, Africa, Globe
D. Globe, Africa, Ethiopia, Adama
3. *"Climate change becomes the serious problem of our globe. Because according to Professor Helen, who is the biodiversity expert, argues that global warming is attacking variety of plants and animals in the world."* This argument is:
A. Inductive- based authority
B. deductive- based on analogy
C. inductive –based on sign
D. deductive- definition
4. *"Ethiopia has ten regional states. The ten regional states have regional state council. Gambela is one of the ten regional states. Therefore, Gambela has a regional state council."* This argument is:
A. cogent inductive argument
B. uncogent inductive argument
C. unsound deductive argument
D. sound deductive argument

5. "In this year, the university registered 1000 regular students and the university provides 700 birr pocket money for each student per month. From this we may conclude that, the university has been spending 700,000birr per month for students' pocket money." This argument is:
- weak inductive argument
 - Valid deductive argument
 - Invalid deductive argument
 - strong Inductive argument
6. 'Russia has promising economy, nuclear energy, and known for its dictator leadership. USA like Russia has promising economy and nuclear energy. Therefore, USA probably also known for its dictator leadership.' This argument is:
- Argument based on statistics
 - Argument based on casual inferences
 - Argument based on analogy
 - Argument based on definition
7. Which one of the following is an argumentative passage?
- 'All USA presidents are born in England. Emmanuel Macron is a USA president. It follows that; Emmanuel Macron is born in England.'
 - 'Ethiopian Broadcasting Cooperation (EBC) television announces that the ministry of education shows progress.'
 - 'If you insert litmus paper to acid, then its color will be changed to red.'
 - "Ethiopia has a wealth of culture. Look the wedding ceremony, and gorgeous celebrations of religious festivals."
8. **Read the following argument!**
'Last year, an international NGO working on youth's behavior collected data from eight hundred regular students from different private colleges in Adama town. The result revealed that five hundred students have been practicing cannabis. Thus, it is reasonable to conclude that, more than seventy percent of regular students from different private colleges in Adama town are practicing cannabis.' Which one of the following is true about the argument stated above?
- strong inductive argument
 - valid deductive argument
 - Weak inductive argument
 - Invalid deductive argument
9. Which one of the following is true about the relationship between validity and soundness of deductive arguments?
- If an argument is unsound, then it will be necessarily invalid.
 - If an argument is sound, then it will be necessarily valid.
 - If an argument is valid, then it will be necessarily sound.
 - If an argument is unsound, then it will be necessarily valid.

10. 'All higher educational institutions are center of excellence. Addis Abeba Science and Technology University is a higher educational institution. Therefore, Addis Abeba Science and Technology University is center of excellence.' This argument is _____.
- A. Sound deductive argument
B. Strong inductive argument
C. Good deductive argument
D. Cogent inductive argument

Part III: Short Answer Questions (2pts. each)

A. Instruction: For the following definitions, determine the type of meaning as Intention and Extension with its respective techniques!

1. "Professional person" means a person such as a teacher, an engineer, a doctor, an architect and others.
- _____

2. The term 'Ethics', is derived from the Greek term 'ethos'; 'ethos' refers to character, 'Ethics' is the study of moral character of human being.
- _____

3. 'Daughter' means a female child.
- _____

4. 'Islamic States' means Saudi Arabia, Yemen, Iran, Iraqi and others.
- _____

5. A "potential difference" exists between two conductors if and **only** if a voltmeter shows a reading when connected to the two conductors.
- _____

B. Instruction: Determine the specific or typical form of deductive and inductive argumentation of the following arguments!

1. 'If payment to private hospitals for the medication is restricted, then the health care center will be promising. If health care center will be promising, then citizens easily get access to medical support in governmental hospitals. Therefore, if payment to private hospitals for the medication is restricted, then citizens easily get access to medical support in governmental hospitals.'
- _____

2. 'No text messages are articulate creations. Some love letters are articulate creations. Therefore, some love letters are not text messages.'
- _____

3. 'The rainfall in Addis Ababa has been more than 10 inches every year for the past thirty years. Therefore, the rainfall next year will probably be more than 10 inches.'

4. 'Over the top of that building there is a Red Cross flag. It suggests that, in that building Red Cross medical services are being provided.'

5. 'All universities are educational institutions. All educational institutions are center of excellences. Therefore, all universities are center of excellence.'

Answer Sheet

Name _____ ID NO _____ Section _____

Part I: True/False

1. _____ 2. _____ 3. _____ 4. _____ 5. _____

Part II: Choose

1. _____ 2. _____ 3. _____ 4. _____ 5. _____

6. _____ 7. _____ 8. _____ 9. _____ 10. _____